Primer: MPEC for Dynamic Discrete Choice

Su and Judd (2012)

econirl package documentation

1 Context and Problem

Structural estimation of DDC models requires solving the Bellman equation at every candidate parameter vector during optimization. In Rust's NFXP approach (Rust, 1987), an outer loop searches over θ while an inner loop solves $V = T(V; \theta)$ by contraction. When the discount factor β is close to 1, the contraction rate β approaches unity and the inner loop requires many iterations. Su and Judd (2012) propose MPEC (Mathematical Programming with Equilibrium Constraints) as an alternative: instead of solving the Bellman equation inside the likelihood evaluation, treat V as a decision variable and enforce the Bellman equation as a constraint. This eliminates the inner fixed-point loop entirely and replaces it with a single constrained optimization over (θ, V) jointly.

The practical advantage is that standard nonlinear programming solvers (KNITRO, SNOPT, or augmented Lagrangian methods) can handle the constraint efficiently, especially when the state space is large and the inner loop is expensive. Iskhakov et al. (2016) later showed that if the NFXP inner solver uses Newton-Kantorovich (NK) iteration instead of successive approximation (SA), the speed advantage of MPEC largely disappears at moderate state spaces ($|\mathcal{S}| < 200$). MPEC remains relevant for very large state spaces and as a conceptual alternative that makes the computational structure explicit.

2 Model and Notation

The DDC model follows Rust (1987). The agent observes state $s_t \in \mathcal{S}$ with $|\mathcal{S}| = n$, chooses action $a_t \in \mathcal{A}$ with $|\mathcal{A}| = k$, and receives flow utility $u(s_t, a_t; \theta) + \sigma\epsilon_{t,a}$ where ϵ is Type I extreme value. The state transitions according to $P(s'|s, a)$ and future payoffs are discounted at rate β.

Table 1: Common DDC notation shared across all primers.

Symbol	Name	Definition
$s \in \mathcal{S}$	State	Discrete or continuous
$a \in \mathcal{A}$	Action	Finite: $\lvert\mathcal{A}\rvert = J$
s'	Next state	Drawn from $P(s' \mid s, a)$
$\beta \in [0, 1)$	Discount factor	
$\sigma > 0$	Scale parameter	Logit noise scale ($= 1$ by convention)
$\theta \in \Theta \subset \mathbb{R}^{d_\theta}$	Structural parameters	Unknown
$\phi(s, a)$	Feature vector	Known functions of (s, a), dimension K
$u(s, a; \theta)$	Flow utility	$\theta^\top \phi(s, a)$ (linear-in-parameters)
$P(s' \mid s, a)$	Transition kernel	Known or estimated from data
$Q(s, a)$	Choice-specific value	$u(s, a; \theta) + \beta \sum_{s'} P(s'\lvert s, a)\, V(s')$
$V(s)$	Integrated value	$\sigma \log \sum_a \exp(Q(s, a)/\sigma)$
$\pi(a \mid s)$	Choice probability	$\exp\big((Q(s, a) - V(s))/\sigma\big)$
τ	Trajectory	$(s_0, a_0, s_1, a_1, \ldots)$ from one agent
$\mathcal{D}$	Panel data	Collection of N observed trajectories

MPEC introduces two additional objects: $\lambda \in \mathbb{R}^n$ (Lagrange multipliers for the Bellman constraint) and $\rho > 0$ (augmented Lagrangian penalty weight). The Bellman operator is $T_s(V; \theta) = \sigma \log \sum_a \exp[(u(s, a; \theta) + \beta P_a V)/\sigma]$.

3 Formulation

NFXP (nested).

$$\max_\theta \ \mathcal{L}(\theta) \quad \text{where } V(\theta) \text{ solves } V = T(V; \theta). \tag{1}$$

The log-likelihood is $\mathcal{L}(\theta) = \sum_{i,t} \log \pi(a_{it} | s_{it}; \theta, V(\theta))$. At each θ, the inner loop iterates $V^{k+1} = T(V^k; \theta)$ until convergence.

MPEC (joint).

$$\max_{\theta, V} \ \mathcal{L}(\theta, V) \quad \text{subject to} \quad V = T(V; \theta). \tag{2}$$

The decision variables are $(\theta, V) \in \mathbb{R}^{p+n}$. The Bellman equation is an explicit equality constraint rather than an implicit fixed point.

Augmented Lagrangian. MPEC (2) is solved via the augmented Lagrangian

$$\mathcal{L}_\rho(\theta, V, \lambda) = -\mathcal{L}(\theta, V) + \lambda^\top [V - T(V; \theta)] + \frac{\rho}{2} \|V - T(V; \theta)\|^2. \tag{3}$$

The penalty ρ is increased across outer iterations until the constraint $\|V - T(V; \theta)\|_\infty < \epsilon$ is satisfied. Each inner subproblem $\min_{\theta, V} \mathcal{L}_\rho$ is solved by L-BFGS-B.

4 Identification

MPEC and NFXP solve the same MLE problem. At convergence, if $V^* = T(V^*; \theta^*)$, the MPEC log-likelihood equals the NFXP log-likelihood. Standard errors are computed by the same formula: the observed information matrix with the implicit derivative $dV/d\theta$ obtained from $(I - \beta P_\pi)\, dV/d\theta = dT/d\theta$ via the implicit function theorem. Identification requires the same rank condition on the feature matrix $\phi(s, a)$ as NFXP.

5 Algorithm

Algorithm 1 MPEC via Augmented Lagrangian (Su & Judd 2012)

Require: Data panel, transitions P, discount β, initial θ_0
Ensure: MLE parameters θ^*, value function V^*

1: Solve $V_0 = T(V_0; \theta_0)$ by value iteration *// warm start*
2: Set $\lambda_0 = 0$, $\rho = \rho_{\text{init}}$
3: **for** $k = 0, 1, \ldots$ (outer AL iterations) **do**
4: **// Inner: minimize augmented Lagrangian**
5: $(\theta^{k+1}, V^{k+1}) \leftarrow \arg\min_{\theta,V} \ \mathcal{L}_\rho(\theta, V, \lambda^k)$ *// L-BFGS-B*
6: **// Check constraint**
7: $c^{k+1} \leftarrow \|V^{k+1} - T(V^{k+1}; \theta^{k+1})\|_\infty$
8: **if** $c^{k+1} < \epsilon_c$ **then**
9: **break** *// Bellman satisfied*
10: **end if**
11: **// Update multipliers and penalty**
12: $\lambda^{k+1} \leftarrow \lambda^k + \rho \cdot [V^{k+1} - T(V^{k+1}; \theta^{k+1})]$
13: $\rho \leftarrow \min(\rho \cdot \gamma_\rho, \ \rho_{\max})$ *// grow penalty*
14: **end for**
15: Compute SEs via implicit function theorem at (θ^*, V^*)
16: **return** θ^*, V^*, SEs

The warm start (line 1) is critical: initializing V_0 from the Bellman solution at θ_0 gives L-BFGS-B a feasible starting point. The penalty schedule $\rho_{\text{init}} = 1$, $\gamma_\rho = 10$, $\rho_{\max} = 10^6$ works well across discount factors.

6 Results from the Paper

Su & Judd (2012, Table 2). On the Rust bus engine with $n = 175$ states and $\beta = 0.9999$, MPEC (using KNITRO) solves in 0.5 seconds versus 188 seconds for NFXP-SA. The speedup comes from avoiding the slow contraction mapping: at $\beta = 0.9999$ the contraction rate is 0.9999 and SA needs roughly $\log(10^{-10})/\log(0.9999) \approx 230{,}000$ inner iterations per outer step.

Iskhakov, Rust & Schjerning (2016, Table 1). When the NFXP inner solver switches from SA to Newton-Kantorovich (NK), NFXP-NK solves the same problem in 0.3 seconds, eliminating MPEC's speed advantage. NK converges in 5–10 inner iterations regardless of β because it achieves quadratic convergence near the fixed point. The speed advantage of MPEC only reappears at very large state spaces ($n > 500$) where forming and inverting the $n \times n$ Jacobian for NK becomes expensive.

Takeaway. The Su & Judd critique of NFXP was correct for NFXP-SA but not for NFXP-NK. The econirl library implements the NK inner solver via the hybrid SA→NK polyalgorithm of Iskhakov et al. (2016), which starts with SA iterations and switches to NK once the contraction residual is small enough for quadratic convergence.

7 Simulation

We run MPEC and NFXP-NK on the Rust bus engine ($n = 90$ mileage bins, true $\theta_c = 0.001$, $RC = 3.0$) at three discount factors. Expert data: 500 buses observed for 100 periods each (50,000 observations). Run `python mpec_run.py` to regenerate Table 2.

β	NFXP-NK			MPEC (Augmented Lagrangian)			
	$\hat{\theta}_c$	$\widehat{RC}$	Time	$\hat{\theta}_c$	$\widehat{RC}$	Time	CV
0.90	0.001191	3.0147	3.0s	0.002771	3.1337	75.1s	$1.1e-05$
0.95	0.001044	3.0079	1.4s	0.001172	3.0225	70.4s	$1.4e-06$
0.99	0.001003	3.0081	4.3s	−0.000253	2.5998	73.1s	$1.5e-04$

Table 2: Parameter recovery on Rust bus (90 states). True: $\theta_c = 0.001$, $RC = 3.0$. CV = Bellman constraint violation $\|V - T(V;\theta)\|_\infty$. Auto-generated by `mpec_run.py`.

NFXP-NK recovers the true parameters within sampling variability at all three discount factors, with OC estimates within 20% of the true 0.001 and RC estimates within 0.5% of the true 3.0. MPEC performs well at $\beta = 0.95$ (OC within 17%, RC within 1%) but shows larger deviations at the extremes. At $\beta = 0.90$ the operating cost is overestimated because the low discount factor makes the Bellman constraint less binding and the AL penalty schedule needs more outer iterations. At $\beta = 0.99$ the constraint is tighter but the landscape is flatter, making L-BFGS-B convergence slower.

The timing comparison confirms the Iskhakov et al. finding: NFXP-NK is substantially faster at this state space size (90 states). The NK inner loop exploits the Jacobian structure of the Bellman operator to achieve quadratic convergence in 5–10 iterations, while the MPEC AL solver must optimize over 92 variables ($p + n = 2 + 90$) at each outer step. MPEC's advantage would emerge at $n > 500$ states where forming the NK Jacobian becomes expensive.

The Bellman constraint violation (CV) measures how well the final V^* satisfies $V = T(V;\theta)$. Values below 10^{-4} indicate the augmented Lagrangian successfully enforced the equilibrium constraint. When CV is larger, the AL penalty ρ did not grow fast enough to enforce feasibility, and the parameter estimates may be biased.

Code mapping. The econirl implementation at `src/econirl/estimation/mpec.py` provides `MPECEstimator` with `MPECConfig`. Key config parameters: `solver="augmented_lagrangian"`, `rho_initial=1.0`, `rho_growth=10.0`, `rho_max=1e6`, `constraint_tol=1e-8`. Standard errors use the same implicit function theorem formula as NFXP because the Bellman constraint is exactly satisfied at convergence.

References

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